



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 11 2009 ASSESSMENT TASK 4

MATHEMATICS/MATHEMATICS EXTENSION 1
YEARLY EXAMINATION

Monday 31st August, 2009

8:30 am to 11:00am

Time Allowed – Two and a half hours plus 5 minutes reading time

WEIGHTING 40% towards final assessment

Outcomes referred to: P2, P3, P4, P5, P6, P7, P8

INSTRUCTIONS:

- 1. Attempt ALL questions.**
- 2. Show all necessary working in every question.**
- 3. Begin each question on a separate piece of paper.**
- 4. There are 10 questions worth 10 marks each. Total marks 100.**
- 5. Marks for individual part questions are shown beside each part.**
- 6. Non- programmable silent Board of studies approved calculators are permitted.**
- 7. Write your name, class and your teacher's name on the front of each answer page.**

Question 1 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

	Marks
(a) Evaluate $\frac{1}{15+5 \times 3}$, correct to three significant figures.	2
(b) Simplify $\sqrt{32} + 4\sqrt{8}$	1
(c) Solve for x : $2^x = \frac{1}{2}$	1
(d) Rationalise the denominator: $\frac{2}{3-\sqrt{5}}$	2
(e) Solve $ x - 3 < 4$	2
(f) Factorise: $9 - 16t^2$	2

Question 2 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

(a) Solve: $\frac{x}{2} + \frac{x}{3} = 1$

2

(b) Simplify: $\frac{x}{x^2-9} - \frac{3}{x-3}$

2

(c) The function $f(x)$ is defined as:

$$f(x) = \begin{cases} 2x & -4 \leq x < 0 \\ 9 - x^2 & 0 \leq x \leq 3 \end{cases}$$

(i) Find the value of $f(0)$.

1

(ii) Sketch $y = f(x)$

2

(iii) State the range of $y = f(x)$

1

(d) Sketch on separate diagrams, showing all essential features:

(i) $y = |x| + 2$

1

(ii) $(x - 2)^2 + y^2 = 9$

1

Question 3 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

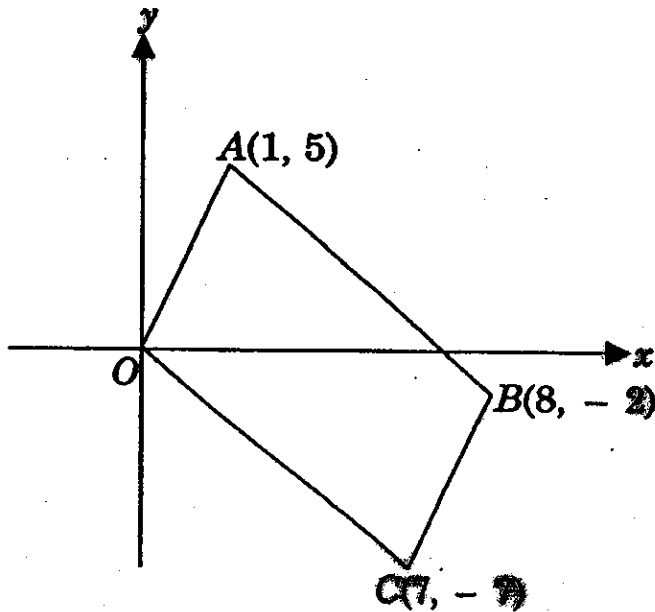


Diagram is NOT drawn to scale

In the diagram, OABC is a quadrilateral where the coordinates of O, A, B and C are $(0,0)$, $(1,5)$, $(8,-2)$ and $(7,-7)$ respectively.

- | | | |
|-----|--|---|
| (a) | Find the midpoint of the interval joining AC. | 1 |
| (b) | Find the gradient of AB. | 1 |
| (c) | Show that the equation of the line containing AB is $x + y = 6$. | 1 |
| (d) | Find the exact length of AB. | 2 |
| (e) | Show that OC is parallel to AB. | 1 |
| (f) | Explain why OABC is a parallelogram. | 1 |
| (g) | Find the perpendicular distance from O to AB. Leave your answer in simplest surd form. | 2 |
| (h) | Find the area of parallelogram OABC. | 1 |

Question 4 (10 marks)

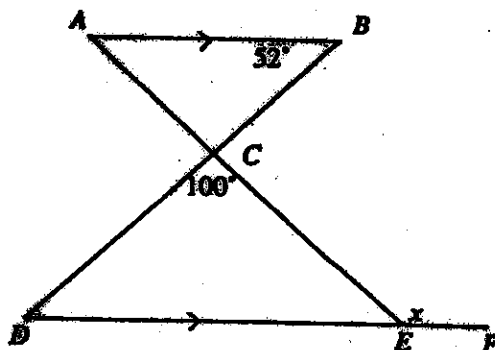
Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

- (a) Solve simultaneously: $3x - y = 5$
 $5x + 3y = -8$

3

(b)



$AB \parallel DE, \angle ABC = 52^\circ, \angle DCE = 100^\circ$.

- (i) Copy the diagram onto your answer page.
 (ii) Find x° giving reasons.

2

- (c) Evaluate: $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

2

- (d) For the parabola $x^2 + 6x - 8y + 9 = 0$, find the:

- (i) coordinates of the vertex.
 (ii) focal length
 (iii) coordinates of the focus

1

1

1

Question 5 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

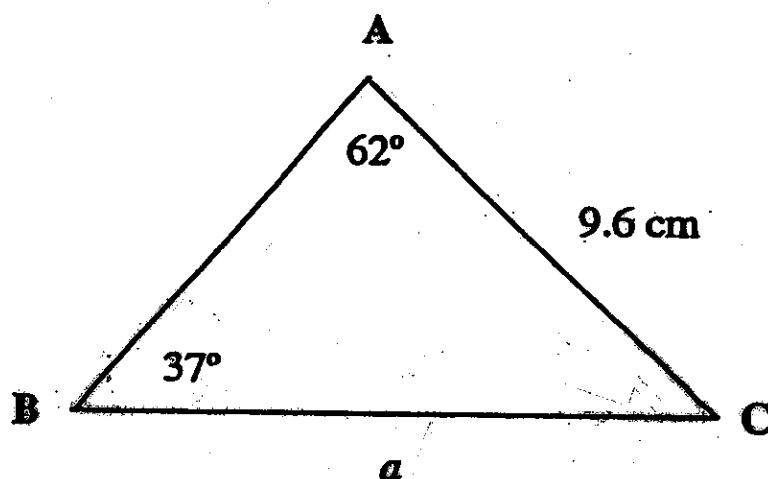
Marks

- (a) Write down the values of θ , $0 \leq \theta \leq 360^\circ$, for which:

$$2 \cos \theta = -1$$

2

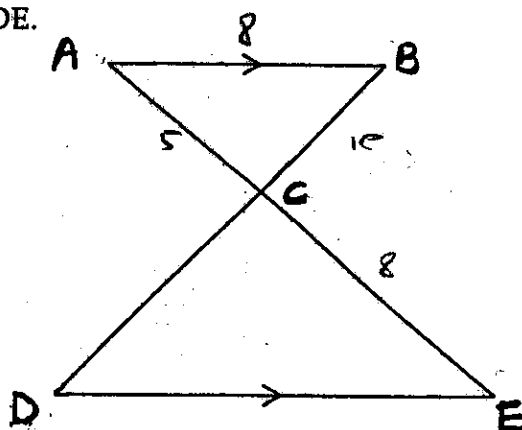
- (b)



In the given triangle, find the length of side a correct to one decimal place.

2

- (c) In the diagram, $AB \parallel DE$.



- (i) Prove that $\triangle ABC$ is similar to $\triangle EDC$.
 (ii) If $AC = 5\text{ cm}$, $AB = 8\text{ cm}$, $BC = 10\text{ cm}$, $CE = 8\text{ cm}$, find the length CD .

3

2

- (d) Evaluate the exact value of: $\sec(-45^\circ)$

1

Question 6 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

	Marks
(a) Find the coordinates of the vertex of $y = x^2 + 4x - 12$	2
(b) Express $7x^2 - 5x + 4$ in the form $Ax(x + 1) + B(x + 1) + C$	3
(c) For what values of k does $2x^2 - 5x + k = 0$ have real roots?	2
(d) Given that α and β are the roots of the quadratic equation $2x^2 - 3x - 5 = 0$, find:	
(i) $\alpha + \beta$	1
(ii) $\alpha\beta$	1
(iii) $(\alpha + 1)(\beta + 1)$	1

Question 7 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

- (a) Indicate clearly on a diagram the region where $y > x + 2$ and $y \geq 3 - x^2$ hold simultaneously. 2
- (b) From a point P , point Q bears 065° and is $20m$ away. Another point R , is $28m$ from P on a bearing of 287° .
- (i) Show this information on a diagram. 1
- (ii) Find the distance (to the nearest metre from Q to R). 3
- (c) Determine whether the function $f(x) = \frac{x^2-1}{x}$ is even, odd or neither. 2
- (d) Factorise fully: $ab - 5b + 20 - 4a$ 2

Question 8 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

(a) Differentiate with respect to x :

(i) $5x^2 - 3x + 1$ 1

(ii) $f(x) = (3x - 1)^{-2}$ 2

(iii) $x(x + 5)^4$ 2

(iv) $y = \frac{x}{(x+3)^2}$ 2

(b) Find the equation of the tangent to the curve $y = 3 - x^4$ at the point where $x = -1$. 3

Question 9 (10 marks)

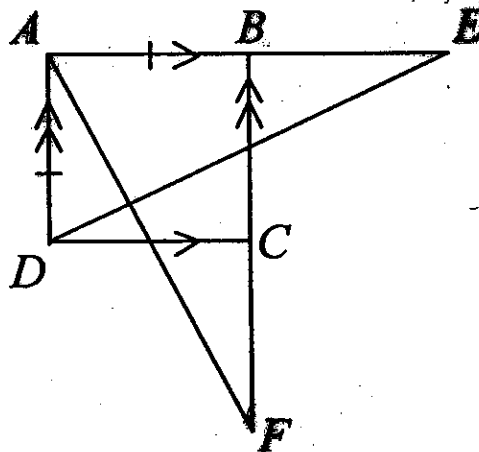
Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

- (a) The equation $4x^2 - px + q = 0$ has one root three times the other. If the smaller is α , show that:

- | | | |
|-------|------------------|---|
| (i) | $p = 16\alpha$ | 2 |
| (ii) | $q = 12\alpha^2$ | 2 |
| (iii) | $3p^2 = 64q$ | 2 |

- (b) In diagram $ABCD$ is a square. AB is produced to E so that $AB = BE$ and BC is produced to F so that $BC = CF$.



- | | | |
|-------|--|---|
| (i) | Copy the diagram onto your answer page. | |
| (ii) | Prove $\triangle AED \equiv \triangle BFA$ | 3 |
| (iii) | Hence prove that $\angle AED = \angle BFA$ | 1 |

Turn over the page for Question 10

Question 10 (10 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

- (a) Solve: $4^x - 12(2^x) + 32 = 0$ 3
- (b) Find all the values of x , $0 \leq x \leq 360^\circ$, for which: $\sin^2 x = \frac{3}{4}$ 3
- (c) (i) The sum of the radii of two circles is 100cm . If one of the circles has a radius $x\text{cm}$, show that the sum of the areas of the two circles is given by
- $$A = 2\pi(x^2 - 100x + 5000)$$
- 3
- (ii) For what values of x is this area a minimum? 1

End of Examination



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QUESTION 1

$$(a) \quad 0.0\dot{3} \quad (1) \\ = 0.0333 \quad (1) \quad (3 \text{ sig figs}).$$

$$(b) \quad 4\sqrt{2} + 8\sqrt{2} \\ = 12\sqrt{2} \quad (1)$$

$$(c) \quad 2^x = 2^{-1} \\ x = -1 \quad (1)$$

$$(d) \quad \frac{2}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} \quad (1) \\ = \frac{2(3 + \sqrt{5})}{9 - 5} \\ = \frac{6 + 2\sqrt{5}}{4} \quad \swarrow \text{up to here was also accepted.} \\ = \frac{3 + \sqrt{5}}{2} \quad (1)$$

$$(e) \quad x - 3 < 4 \quad \text{OR} \quad -x + 3 < 4 \\ x < 7 \quad (1) \quad \quad \quad -x < 1 \\ \quad \quad \quad \quad \quad \quad \quad \quad x > -1 \quad (1)$$

$$(f) \quad (3 - 4t)(3 + 4t) \\ \quad \quad (1) \quad \quad (1)$$

QUESTION 2

$$(a) \quad \frac{3x + 2x}{6} = 1 \quad (1)$$

$$5x = 6$$

$$x = \frac{6}{5} \text{ or } 1\frac{1}{5} \quad (1)$$

$$(b) \quad \frac{x}{(x-3)(x+3)} - \frac{3}{x-3} \quad (1)$$

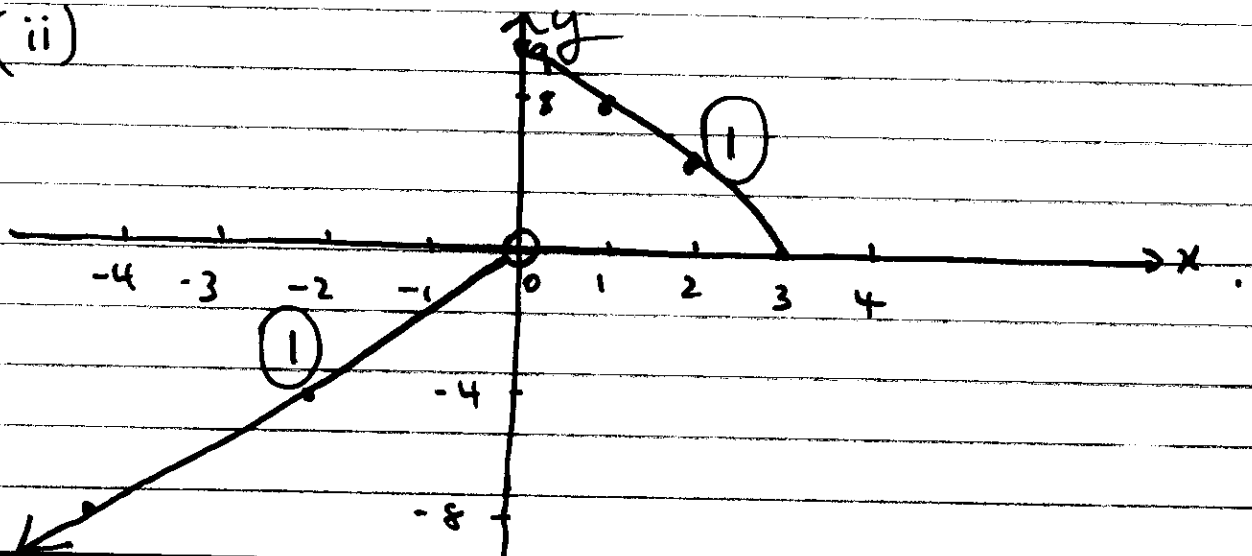
$$= \frac{x - 3(x+3)}{(x-3)(x+3)}$$

$$= \frac{x - 3x - 9}{(x-3)(x+3)}$$

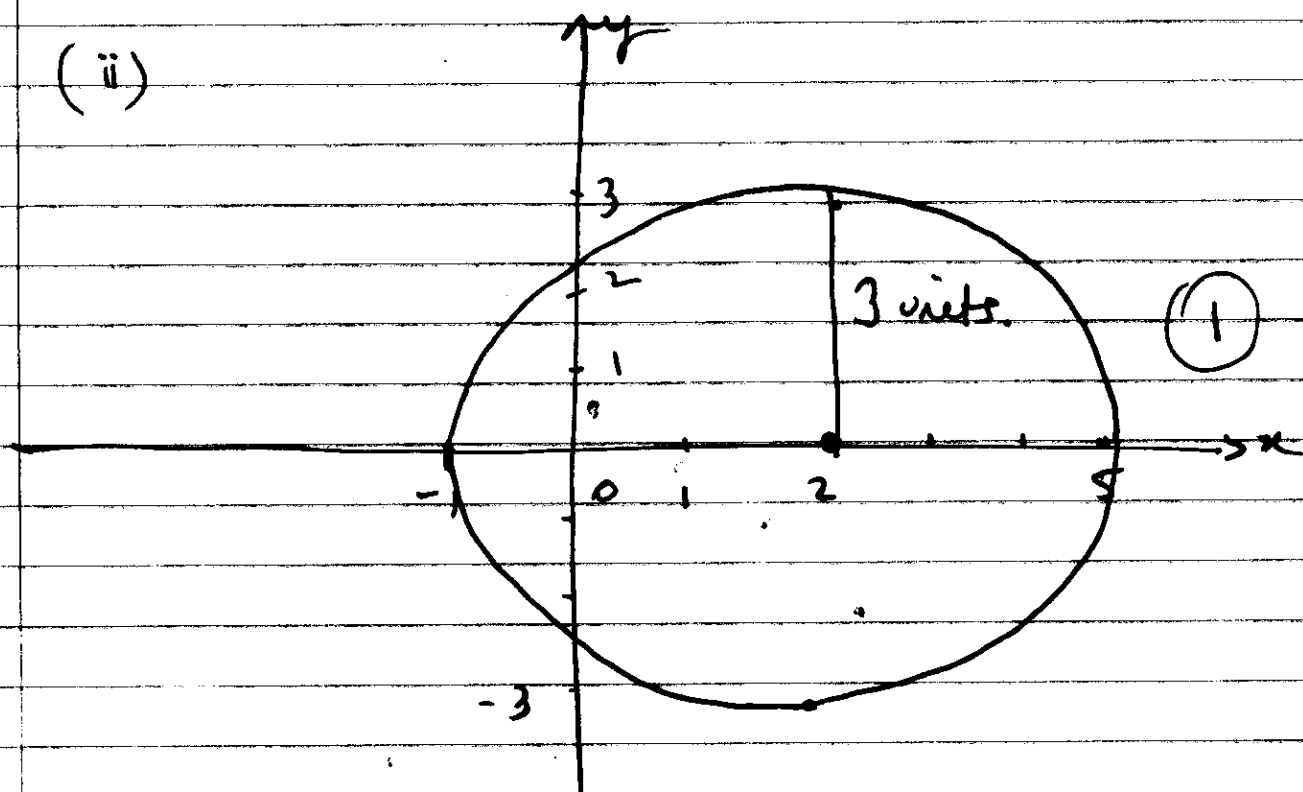
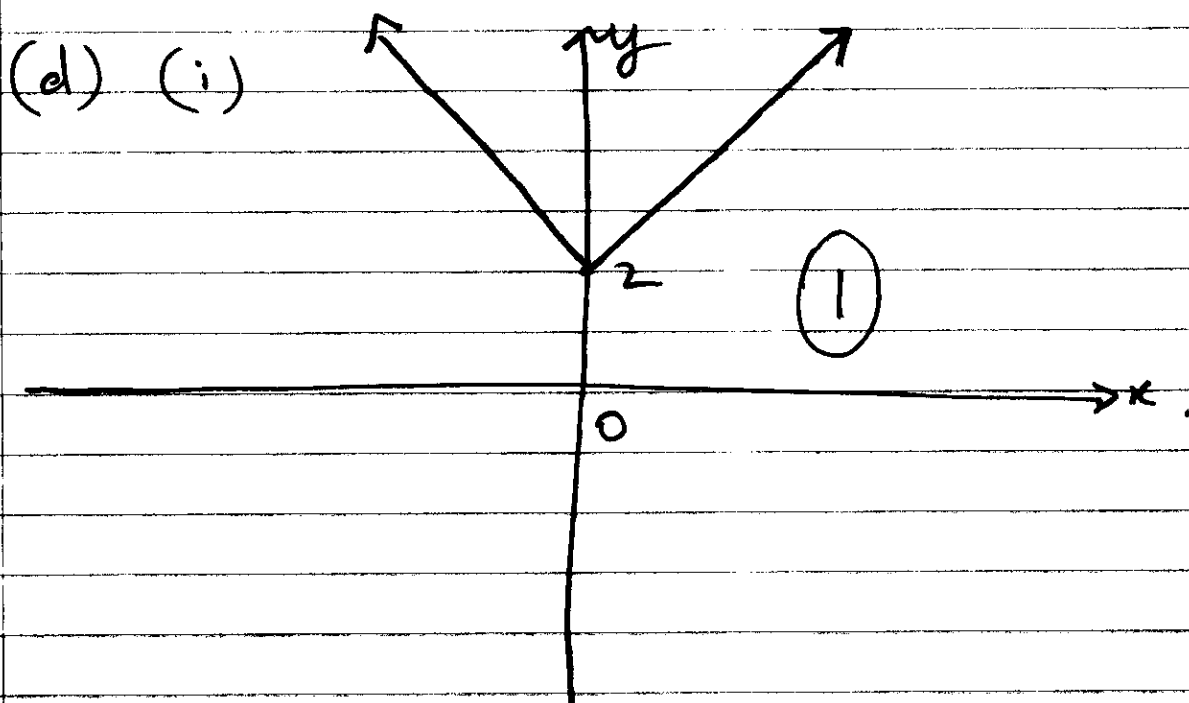
$$= \frac{-2x - 9}{(x-3)(x+3)} \quad (1)$$

$$(c) \quad (i) \quad f(0) = 9 - (0)^2 = 9 \quad (1)$$

(ii)



(iii) $R: \{-8 \leq y \leq 9\}$ (1)



QUESTION 3

$$(a) \quad M(x, y) = \left(\frac{1+7}{2}, \frac{5-7}{2} \right) \\ = (4, -1) \quad (1)$$

$$(b) \quad m = \frac{-2-5}{8-1}$$

$$m = \frac{-7}{7}$$

$$m_{AB} = -1 \quad (1)$$

$$(c) \quad y - y_1 = m(x - x_1) \quad m = -1, (1, 5)$$

$$y - 5 = -1(x - 1)$$

$$y - 5 = -x + 1 \quad (1)$$

$$x + y = 6$$

$$(d) \quad d_{AB} = \sqrt{(8-1)^2 + (-2-5)^2} \quad (1)$$

$$= \sqrt{7^2 + 7^2}$$

$$= \sqrt{98} \quad \checkmark \text{ accepted.}$$

$$d_{AB} = 7\sqrt{2} \quad (1) \quad (\text{exact length})!$$

$$(e) \quad m_{OC} = \frac{-7-0}{7-0}$$

$$m_{OC} = -1 = m_{AB} \text{ from (b).} \quad (1)$$

(f) OABC is a parallelogram because

sides $AB \parallel OC$ and $d_{AB} = 7\sqrt{2}$

$$= \sqrt{(7-0)^2 + (-7-0)^2}$$

$$= \sqrt{98} \quad (1)$$

$$\text{doc } 7\sqrt{2}$$

$$(g) \quad d = \frac{|0 \times 1 + 0 \times 1 - 6|}{\sqrt{1^2 + 1^2}} \quad (1) \quad O(0,0)$$

$$x + y = 6$$

$$d = \frac{|-6|}{\sqrt{2}}$$

$$d = \frac{6}{\sqrt{2}} \quad \text{or} \quad \frac{6\sqrt{2}}{2} = 3\sqrt{2} \text{ units.} \quad (1)$$

(h) Area = base $\times$ perpendicular height.

$$\text{Area} = 7\sqrt{2} \times 3\sqrt{2}$$

$$= 21 \times 2$$

$$\text{Area} = 42 \text{ unit}^2 \quad (1)$$

QUESTION 4

(a) $9x - 3y = 15$ — (1)

$5x + 3y = -8$ — (2)

(1) + (2)

$14x = 7$

$x = \frac{1}{2}$ (1)

sub x into (1)

$9(\frac{1}{2}) - 3y = 15$

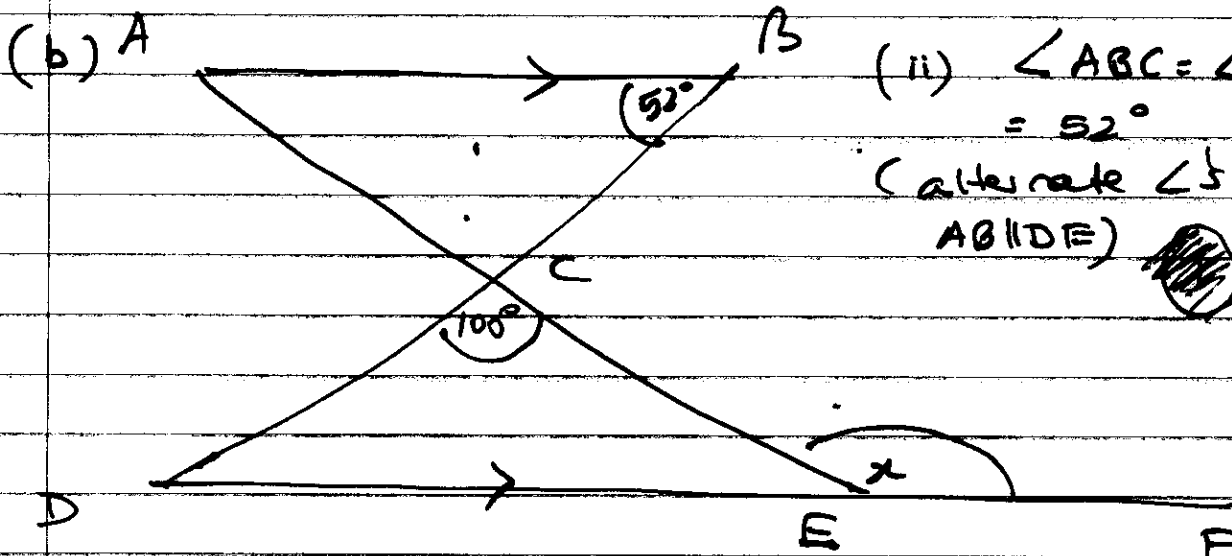
$\frac{9}{2} - 3y = 15$

$-3y = \frac{21}{2}$

$y = -3\frac{1}{2}$ (1)

$(\frac{1}{2}, -3\frac{1}{2})$

* if there was a mistake from the elimination process, (-1) , but then carried through.



In $\triangle CDE$, angle sum of $\triangle = 180^\circ$

$$\therefore 100^\circ + 52^\circ + \angle CED = 180^\circ$$

$$\angle CED = 28^\circ \quad (1)$$

$$\begin{aligned} \therefore \angle CEF = x^\circ &= 180^\circ - 28^\circ \\ &= 152^\circ \end{aligned} \quad (1)$$

straight angle.

$$(c) \quad \lim_{x \rightarrow 2} \frac{(\cancel{x-2})(x^2 + 2x + 4)}{(\cancel{x-2})} \quad (1)$$

$$\lim_{x \rightarrow 2} (x^2 + 2x + 4)$$

$$= 2^2 + 2(2) + 4$$

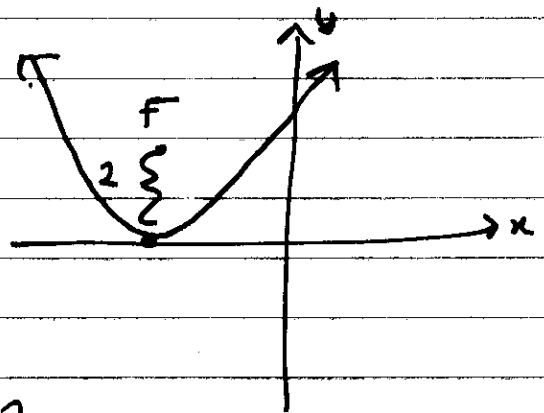
$$= 12 \quad (1)$$

$$(d) \quad x^2 + 6x = 8y - 9$$

$$(x+3)^2 - 9 = 8y - 9$$

$$(x+3)^2 = 8y$$

$$\therefore V(-3, 0) \quad a = 2$$



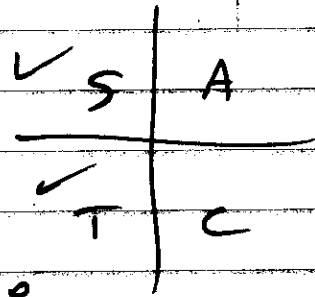
$$(i) \quad V(-3, 0) \quad (1) \quad (iii) \quad F(-3, 2) \quad (1)$$

$$(ii) \quad a = 2 \quad (1) \quad \text{* if mistake with vertex, but the rest carried through correctly (-1)}$$

QUESTION 5

(a) $\cos \theta = -\frac{1}{2}$

$$\cos 60^\circ = \frac{1}{2}$$



$$\therefore \theta = 120^\circ, 240^\circ$$

(1)

(1)

(b) $\frac{a}{\sin 62^\circ} = \frac{9.6}{\sin 37^\circ}$

$$a = \frac{9.6 \times \sin 62^\circ}{\sin 37^\circ} \quad (1)$$

$$a = 14.1 \text{ cm} \quad (1 \text{ dec. pl})$$

(1)

(c) (i) $\angle ACB = \angle ECD$ — vertically opp $\angle$'s are equal.

(1)

$$\angle BAC = \angle CED$$
 — alternate $\angle$'s in $\parallel$ lines are equal.

(1)

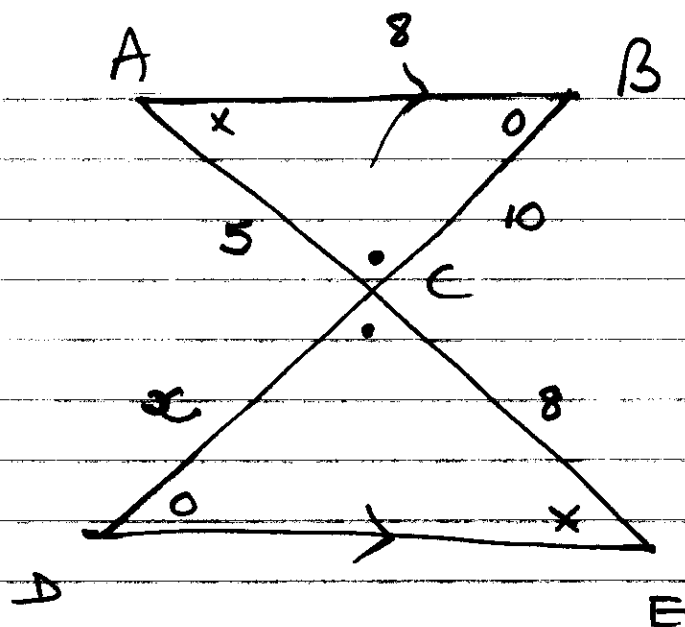
$$\angle ABC = \angle EDC$$
 — alternate $\angle$'s in $\parallel$ lines are equal.

(1)

$\therefore$ equiangular, $\triangle ABC \parallel \triangle EDC$

(ii)

(ii)



Since $\triangle ABC \parallel \triangle EDC$ has been proved.

$$\frac{5}{8} = \frac{10}{x}$$

① corresponding sides are in the same ratio.

$$5x = 80$$

$$x = 16 \text{ cm. } \textcircled{1}$$

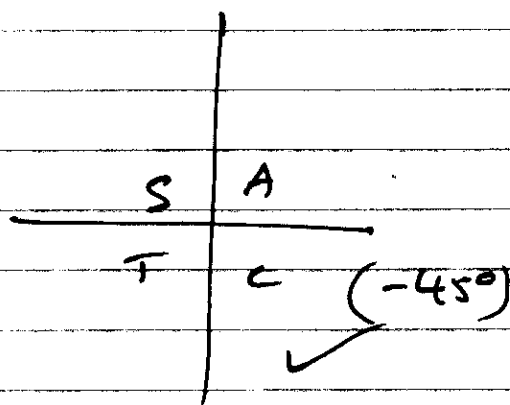
(d)

$$\frac{1}{\cos(-45^\circ)}$$

$$= \frac{1}{(+)\frac{1}{\sqrt{2}}}$$

$$= \sqrt{2}$$

①



QUESTION 6

(a) $x = \frac{-b}{2a}$ (axis of symmetry)

$$x = \frac{-4}{2(1)}$$

$$x = -2 \quad (1)$$

$\therefore$ sub $x = -2$ into y .

$$y = (-2)^2 + 4(-2) - 12$$

$$y = 4 - 8 - 12$$

$$y = -16$$

$$\therefore V(-2, -16) \quad (1)$$

(b) $7x^2 - 5x + 4 \equiv Ax^2 + Ax + Bx + B + C$

$$\equiv Ax^2 + (A+B)x + (B+C)$$

$\therefore$ equating coefficients.

$$\boxed{A = 7} \quad (1)$$

$$A + B = -5$$

$$7 + B = -5$$

$$\boxed{B = -12} \quad (1)$$

$$B + C = 4$$

$$-12 + C = 4$$

$$\boxed{C = 16} \quad (1)$$

$$\therefore 7x^2 - 5x + 4 \equiv 7x(x+1) - 12(x+1) + 16$$

(Students need to express in the above form).

(c) $\Delta \geq 0$ for real roots

$$b^2 - 4ac \geq 0 \quad (1)$$

$$(-5)^2 - 4(2)(k) \geq 0$$

$$25 - 8k \geq 0$$

$$25 \geq 8k$$

$$\frac{25}{8} \geq k$$

$$\therefore k \leq 3\frac{1}{8} \quad (1)$$

(d) (i) $\alpha + \beta = -\frac{b}{a}$

$$= \frac{3}{2} \quad (1)$$

(ii) $\alpha\beta = \frac{c}{a}$

$$= -\frac{5}{2} \quad (1)$$

(iii) $(\alpha+1)(\beta+1)$

$$= \alpha\beta + \beta + \alpha + 1$$

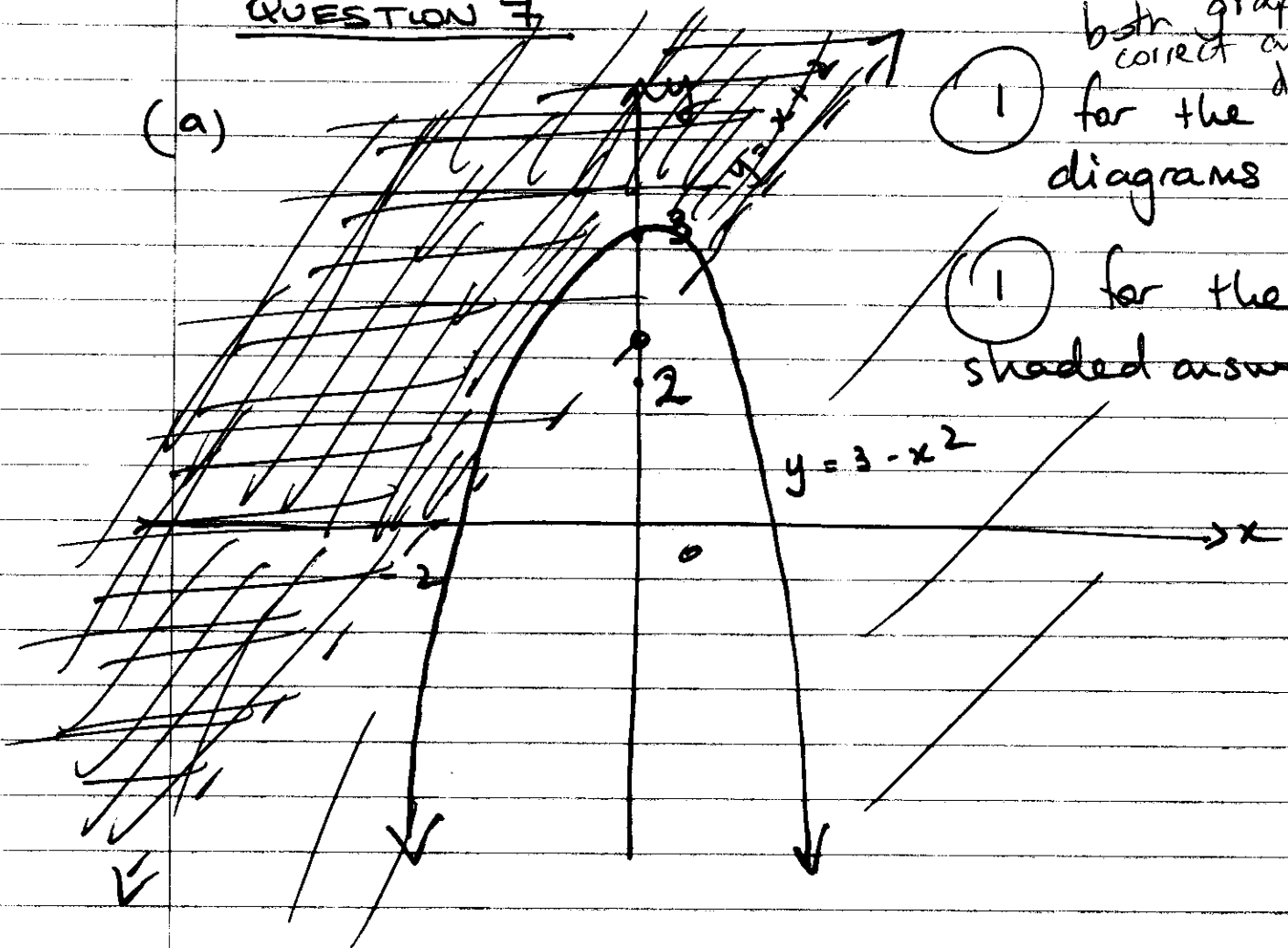
$$= \alpha\beta + (\alpha + \beta) + 1$$

$$= -\frac{5}{2} + \frac{3}{2} + 1$$

$$= 0 \quad (1)$$

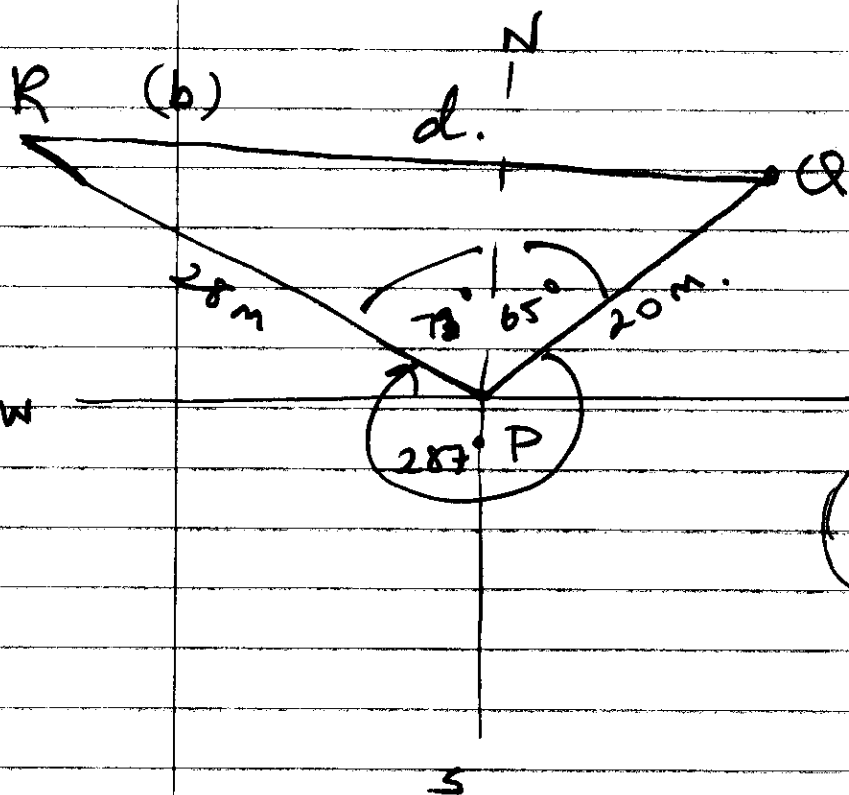
QUESTION 7

(a)



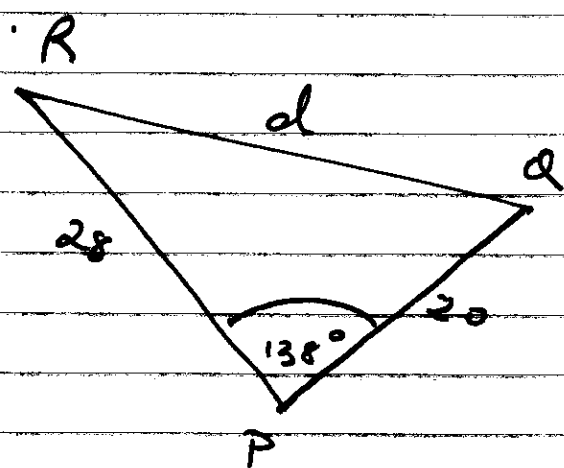
both graphs correct and dotted/ solid
 (1) for the diagrams

(1) for the shaded answer.



(1)

(1)



$$(ii) d_{ar}^2 = 28^2 + 20^2 - 2(28)(20) \cos 138^\circ \quad (1)$$

$$d_{ar}^2 = 2016.32 \quad (1)$$

$$d_{ar} = 45 \text{ m.} \quad (1)$$

↑
if incorrect
can still
get 2
if included L

$$(c) f(x) = \frac{x^2 - 1}{x}$$

$$f(-x) = \frac{(-x)^2 - 1}{-x}$$

$$= \frac{x^2 - 1}{-x}$$

$$= \frac{-(x^2 - 1)}{x}$$

$$f(-x) = \frac{-x^2 + 1}{x}$$

$$\therefore \text{for even } f(x) = f(-x)$$

$$\frac{x^2 - 1}{x} \neq \frac{-x^2 + 1}{x}$$

$$\text{for odd } f(-x) = -f(x)$$

$$\frac{-x^2 + 1}{x} = - \left(\frac{x^2 - 1}{x} \right) \quad (1)$$

$$= \frac{-x^2 + 1}{x} \quad (1)$$

$$f(-x) = -f(x)$$

odd function.

$$\begin{aligned} (d) \quad & b(a-5) + 4(5-a) \\ &= b(a-5) - 4(a-5) \quad (1) \\ &= (b-4)(a-5) \quad (1) \end{aligned}$$

QUESTION 8

$$(a) (i) \frac{d}{dx} = 10x - 3 \quad (1)$$

$$(ii) f'(x) = -2(3x-1)^{-3} \times (3) \quad (1) \\ = -6(3x-1)^{-3}$$

OR

$$f'(x) = \frac{-6}{(3x-1)^3}$$

$$(iii) \frac{d}{dx} = x \times 4(x+5)^3 + 1 \times (x+5)^4 \quad (1) \\ = 4x(x+5)^3 + (x+5)^4 \\ = (x+5)^3 [4x + (x+5)] \\ = (x+5)^3 (5x+5)$$

$$(iv) \frac{dy}{dx} = \frac{(x+3)^2 \times 1 - x[2(x+3)^1]}{(x+3)^4} \quad (1) \\ = \frac{(x+3)^2 - 2x(x+3)}{(x+3)^4} \quad (1)$$

$$= \frac{x^2 + 6x + 9 - 2x^2 - 6x}{(x+3)^4}$$

$$\frac{dy}{dx} = \frac{-x^2 + 9}{(x+3)^4}$$

$$(b) \quad \frac{dy}{dx} = -4x^3 \quad (1)$$

$$\frac{dy}{dx} \text{ at } x = -1$$

$$m = -4(-1)^3$$

$$m = 4 \quad (1)$$

$$\therefore \text{ when } x = -1, y = 3 - (-1)^4$$

$$(-1, 2)$$

$$m = 4$$

$$y = 3 - 1$$

$$y = 2$$

$$y - 2 = 4(x - -1)$$

$$y - 2 = 4x + 4$$

$$y = 4x + 6 \quad (1) \text{ or } 4x - y + 6 = 0.$$

QUESTION 9

(a) (i) let $\alpha, 3\alpha$.

$$\alpha + \beta = \frac{-b}{a}$$

$$\alpha + 3\alpha = \frac{-p}{4} \quad (1)$$

$$4\alpha = \frac{p}{4}$$

~~$$\alpha = \frac{p}{16}$$~~

$$16\alpha = p \quad (1)$$

(ii) $\alpha\beta = \frac{c}{a}$

$$\alpha \times (3\alpha) = \frac{q}{4} \quad (1)$$

$$3\alpha^2 = \frac{q}{4}$$

$$12\alpha^2 = q \quad (1)$$

(iii) if from (i) $\alpha = \frac{p}{16}$

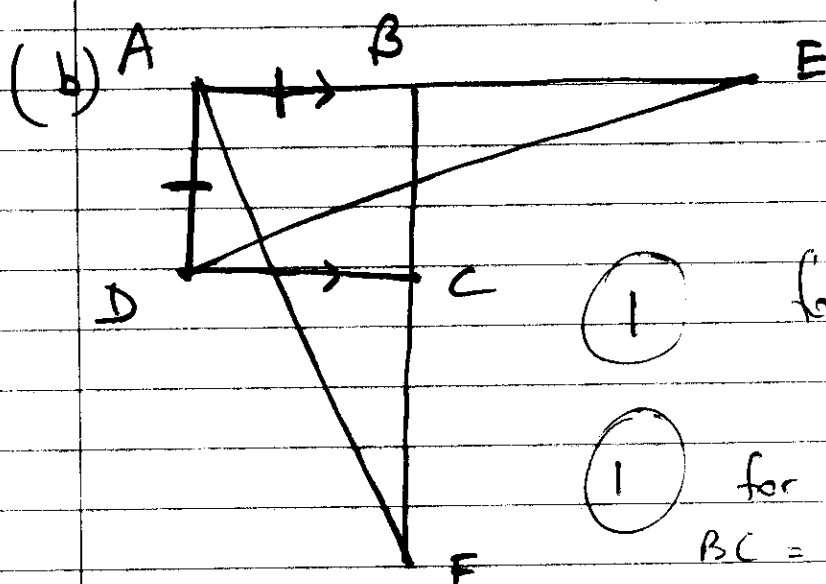
sub α into (ii)

$$12\left(\frac{p}{16}\right)^2 = q \quad (1)$$

$$12 \times \left(\frac{p^2}{256} \right) = 9$$

$$\frac{3p^2}{64} = 9$$

$$3p^2 = 64 \times 9 \quad (1)$$



(1)

for $\angle 90^\circ$ and $AD = AB$.

(1)

for stating $AB = BE$ and $BC = CF$

(ii) ~~AD~~ $AD = AB$ — given (sides of a square are equal).

~~∠~~ $\angle EAD = \angle FBA = 90^\circ$ — interior angles of a square are 90° .

Since $AB = BE$ and $BC = CF$ and

$AB = AD$,

~~∴~~ $AE = BF$

(1)

$\therefore \triangle AED \equiv \triangle BFA$ (SAS)

(iii) Since $\triangle AED \cong \triangle BFA$,

$\angle AED = \angle BFA$, corresponding
angles in congruent $\triangle$'s. (1)

QUESTION 10

$$(a) \quad (2^2)^x - 12(2^x) + 32 = 0$$

$$(2^x)^2 - 12(2^x) + 32 = 0$$

$$\therefore \text{ let } m = 2^x \quad (1)$$

$$m^2 - 12m + 32 = 0$$

$$(m - 4)(m - 8) = 0 \quad (1)$$

$$\therefore m = 4 \text{ or } m = 8$$

$$\text{Since } m = 2^x.$$

$$2^x = 4$$

$$2^x = 2^2$$

$$\boxed{x = 2}$$

(1)

OR

$$2^x = 8$$

$$2^x = 2^3$$

$$\boxed{x = 3}$$

(b)

$$\sin x = \pm \sqrt{\frac{3}{4}}$$

$$\sin x = \pm \frac{\sqrt{3}}{2}$$

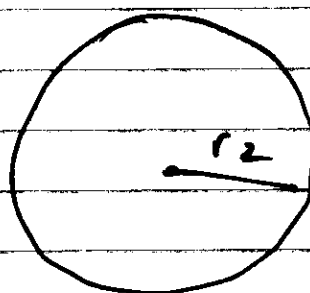
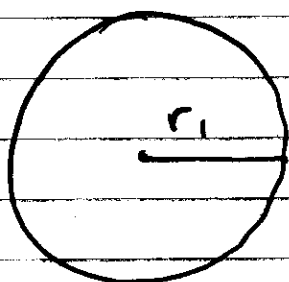
(1)

✓	✓
S	A
✓ T	✓ C

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\therefore x = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

(c)(i)



$$\text{let } r_1 + r_2 = 100 \text{ cm}$$

$$\text{if } r_1 = x$$

$$\therefore x + r_2 = 100$$

$$r_2 = 100 - x \text{ cm.}$$

(1)

$$A_1 = \pi x^2$$

$$A_2 = \pi (100 - x)^2 = \pi (10000 - 200x + x^2)$$

$$\therefore A = A_1 + A_2$$

$$A = \pi x^2 + 10000\pi - 200\pi x + \pi x^2$$

(1)

$$A = 2\pi x^2 - 200\pi x + 10000\pi$$

$$A = 2\pi (x^2 - 100x + 5000)$$

(1)

$$(ii) \quad \frac{dA}{dx} = 2\pi(2x - 100)$$

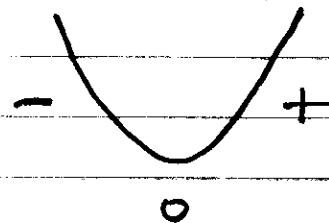
$$\text{let } \frac{dA}{dx} = 0$$

$$0 = 2\pi(2x - 100)$$

$$50 = x$$

(1)

x	40	50	60
$\frac{dA}{dx}$	-	0	+



∴ for $x = 50$, this area is a minimum.

* to get the 1 mark, must test stationary pt.